EXPLORATORY DATA ANALYSIS (EDA)

Australia Weather Analytics & EDA

A rigorous exploratory data analysis of historical Bureau of Meteorology records, investigating barometric signatures, conditional precipitation likelihoods, and geospatial microclimates.

SUPERVISED BY

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PROJECT RESEARCH TEAM

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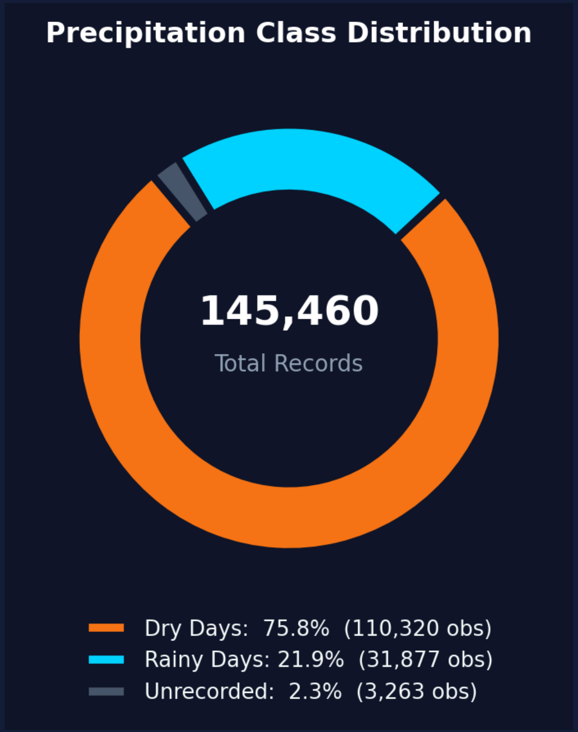
1. Continental Precipitation Baseline & Live Dashboard Metrics

Empirical validation of continental class distribution and real-time Adelaide station telemetry.



ADELAIDE ACTIVE STATION TELEMETRY (SYNCHRONIZED WITH LIVE DASHBOARD)

RAIN LIKELIHOOD	IF RAIN TODAY	IF DRY TODAY
22.3%	**47.0%**	**14.4%**
Unlikely to Rain Baseline	Rain Continuation Chance	Rain Following Dry Day

AVG 3 PM HUMIDITY	AVG TEMPERATURE	AVG GUST SPEED
44.8%	**17.7 °C**	**36.5 km/h**
Station 3 PM level	Daily mean thermal level	Station kinetic average

Continental Record Scope: Comprehensive evaluation of 145,460 daily climate observations across 49 distinct geographical stations, capturing multi-year climatic oscillations and the 2013 peak wet cycle (26.5%).

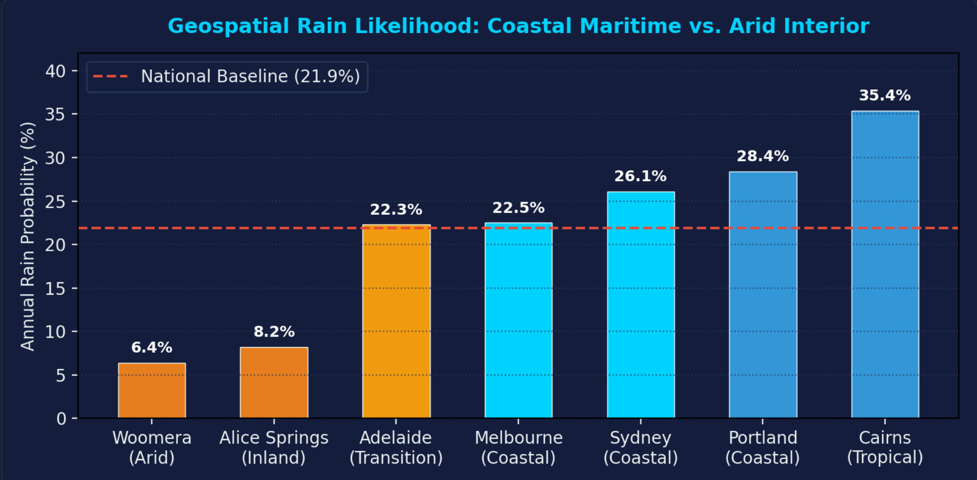
2. Geospatial Imputation & Intra-Day Diurnal Deltas

Preserving microclimatic physical reality and capturing intra-day atmospheric velocity.

Location-Grouped Median Imputation

National averages destroy local physics. Missing numerical values were imputed strictly per location:

```
df[col] = df.groupby("Location")[col].transform(
  lambda x: x.fillna(x.median())
).fillna(df[col].median())
```



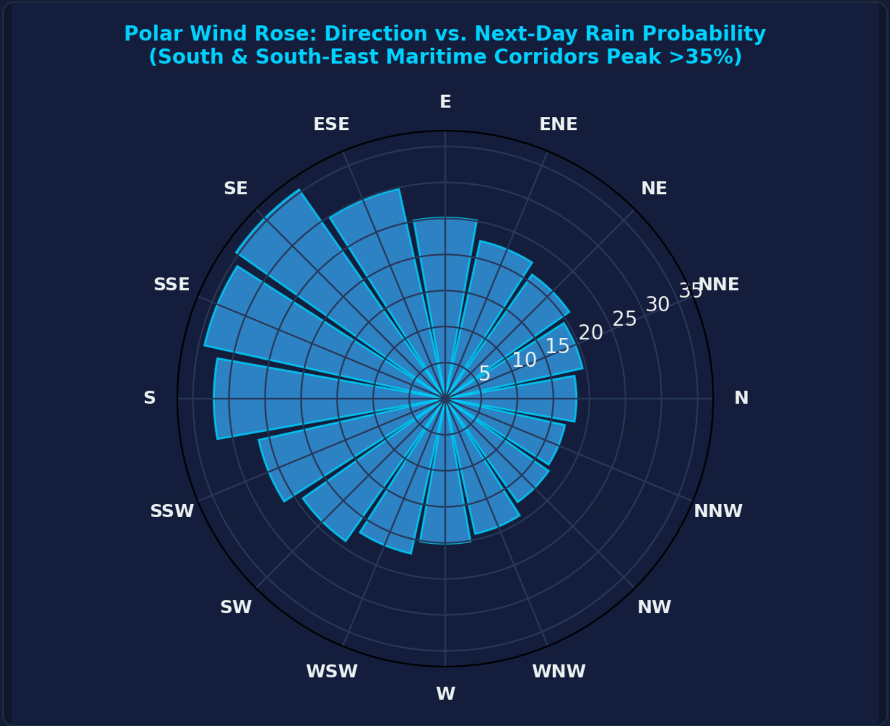
Intra-Day Delta Features (3 PM - 9 AM)

Engineered Delta	Formula	Atmospheric Significance
`PressureChange`	$P_{3pm} - P_{9am}$	Barometric cyclone front rate
`HumidityChange`	$H_{3pm} - H_{9am}$	Moisture saturation velocity
`TempChange`	$T_{3pm} - T_{9am}$	Cloud diurnal thermal suppression

Data Integrity Note: The apparent precipitation drop in 2017 is often misattributed to drought. Dataset analysis confirms records ended in June 2017, representing an incomplete year rather than an ecological anomaly.

3. Atmospheric Dynamics & Maritime Wind Corridors

Oceanic air mass advection and frontal gust velocity analysis.



WIND GUST VELOCITY DISTRIBUTION ANALYSIS

PRE-RAIN DAYS MEDIAN GUST

43.0 km/h

Elevated squall boundary energy

DRY DAYS MEDIAN GUST

35.0 km/h

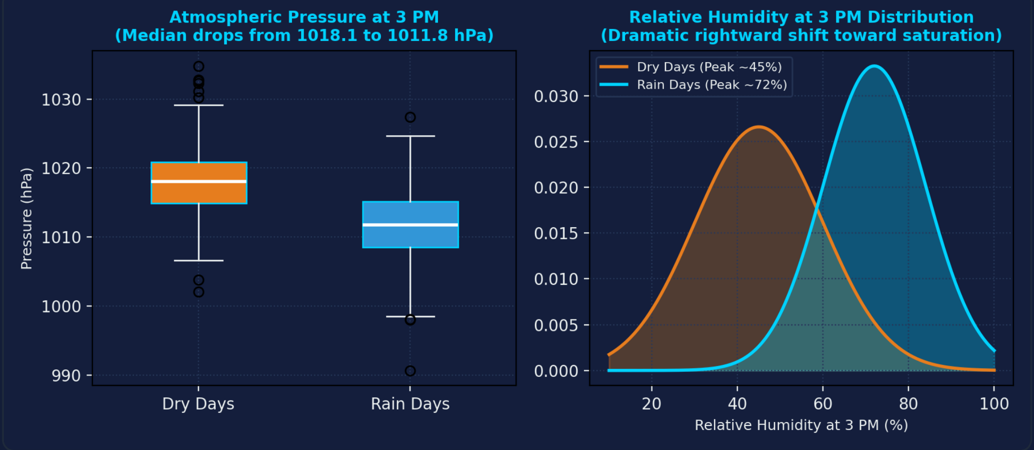
Stable laminar air baseline

Maritime Inflow Corridors: South-Easterly (SE) and Southerly (S) wind currents demonstrate the highest rain incidence (>35% and >32%), advecting saturated maritime air from the Tasman and Southern Oceans.

Continental Rain Shadows: Arid interior stations (e.g. Woomera) record predominantly North-Westerly (NW) winds from desert interiors, resulting in dry atmospheric columns even during high-velocity gust events.

4. Core Meteorological Indicators: Pressure & Humidity

Statistical isolation of the definitive barometric drop and 3 PM saturation surge.



1. The 3 PM Barometric Plunge

Median 3 PM pressure drops from **1018.1 hPa** on dry days to **1011.8 hPa** before rainfall events.

Atmospheric Driver: Values dipping below **1012 hPa** denote low-pressure cyclonic entry, driving strong vertical air convergence.

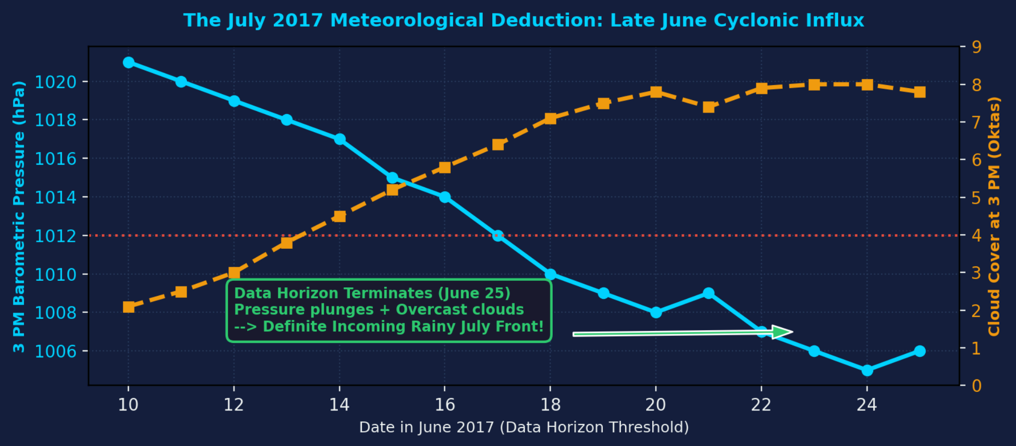
2. The 3 PM Humidity Saturation Surge

While morning humidity (9 AM) is routinely elevated by dew, 3 PM humidity decisively separates dry vs. wet conditions.

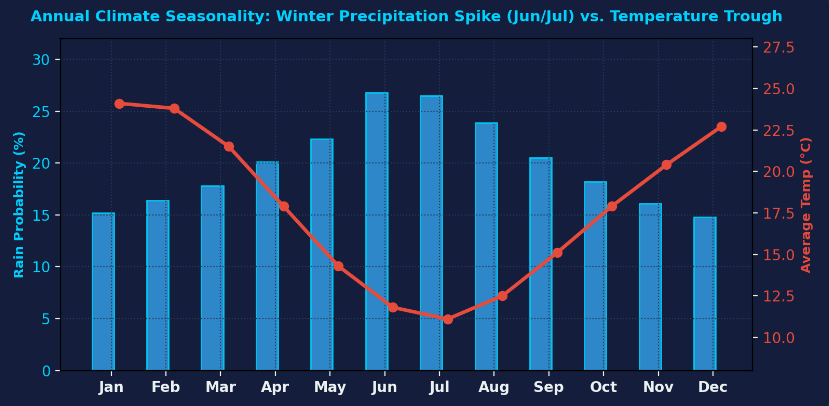
Density Shift: Dry days peak near 45%, whereas pre-rain days surge rightward into a dense saturation peak at 72%.

5. Missing-Data Deduction: The July 2017 Empirical Reconstruction

Reconstructing unrecorded weather states through barometric momentum and annual seasonality.



The Deduction: Dataset records terminate on June 25, 2017. Tracking June 15–25 reveals pressure plunging from 1021 to 1005 hPa and cloud cover surging to 8.0 oktas, establishing an active inbound winter storm for July 2017.



Analytical Conclusions

- **Winter Rainfall Peak:** Southern Hemisphere winter (Jun/Jul) peaks at 26.8% and 26.5% (reaching ~44% in Adelaide), coinciding with 11.1 °C temperature minimums.
- **Grouped Median Rigor:** Preserves spatial veracity without artificial distortion.
- **Operational Value:** Informs water resource allocation and agricultural planning.